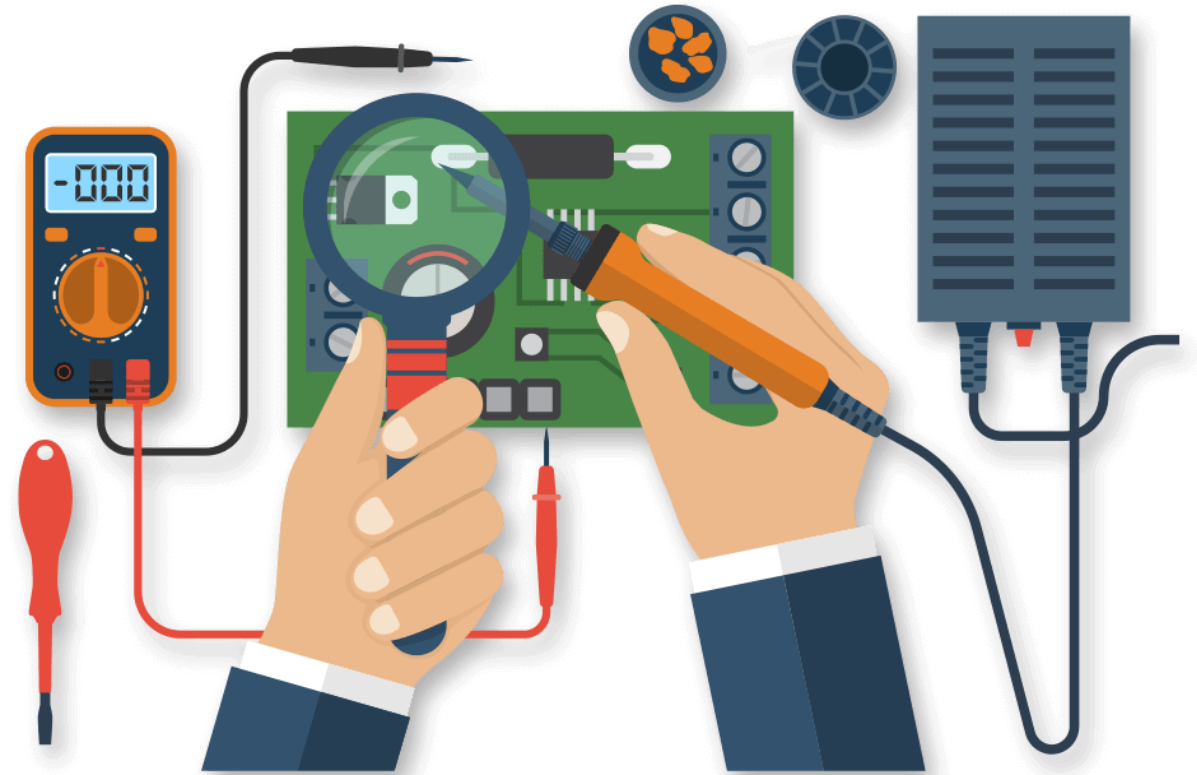


Computer System Architecture

Day 2: DC Circuit Analysis



Supun Chandana Gajendrasinghe

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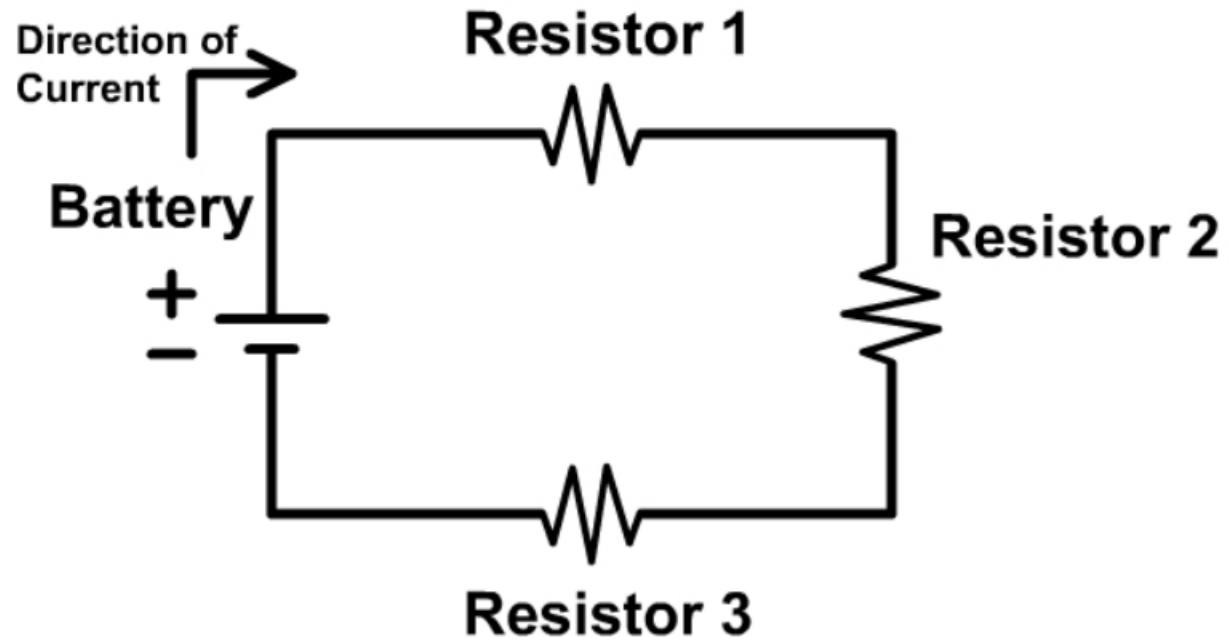
supun.g@nsbm.ac.lk

Outline

- Series Circuits
- Parallel Circuits
- Series Parallel Circuits

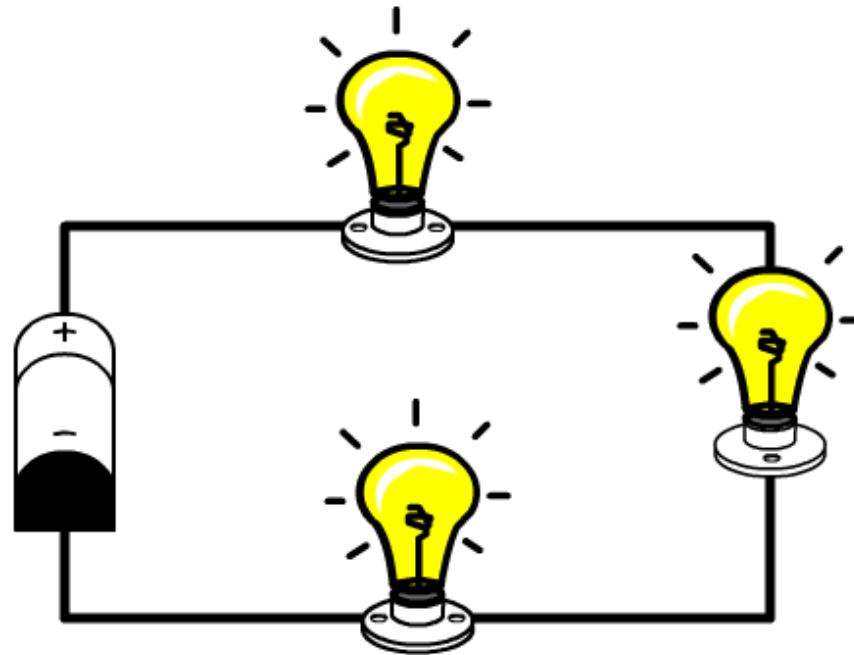


Series Circuits



Series Circuits only have one path. No branches

Series Circuits



Series Circuit

Series Circuits

Series Circuit Rules

$$V_T = V_1 + V_2 + V_3 + \dots$$

$$I_T = I_1 = I_2 = I_3 = \dots$$

$$R_T = R_1 + R_2 + R_3 + \dots$$

Series Circuits



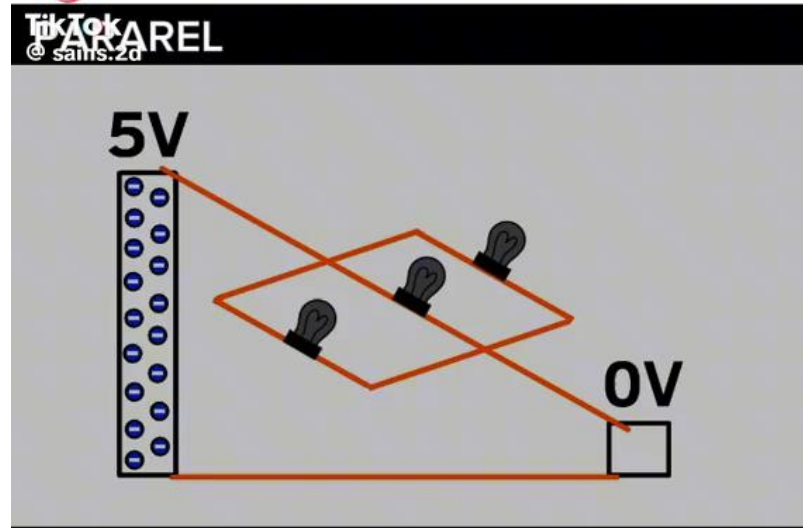
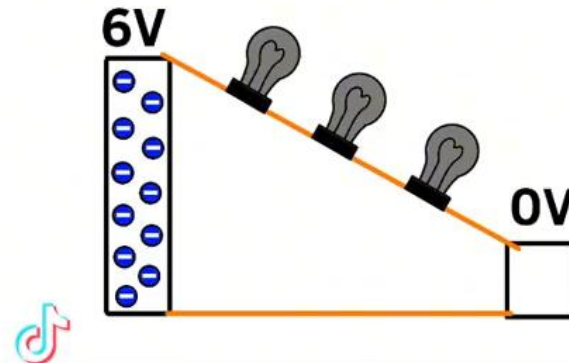
0 V



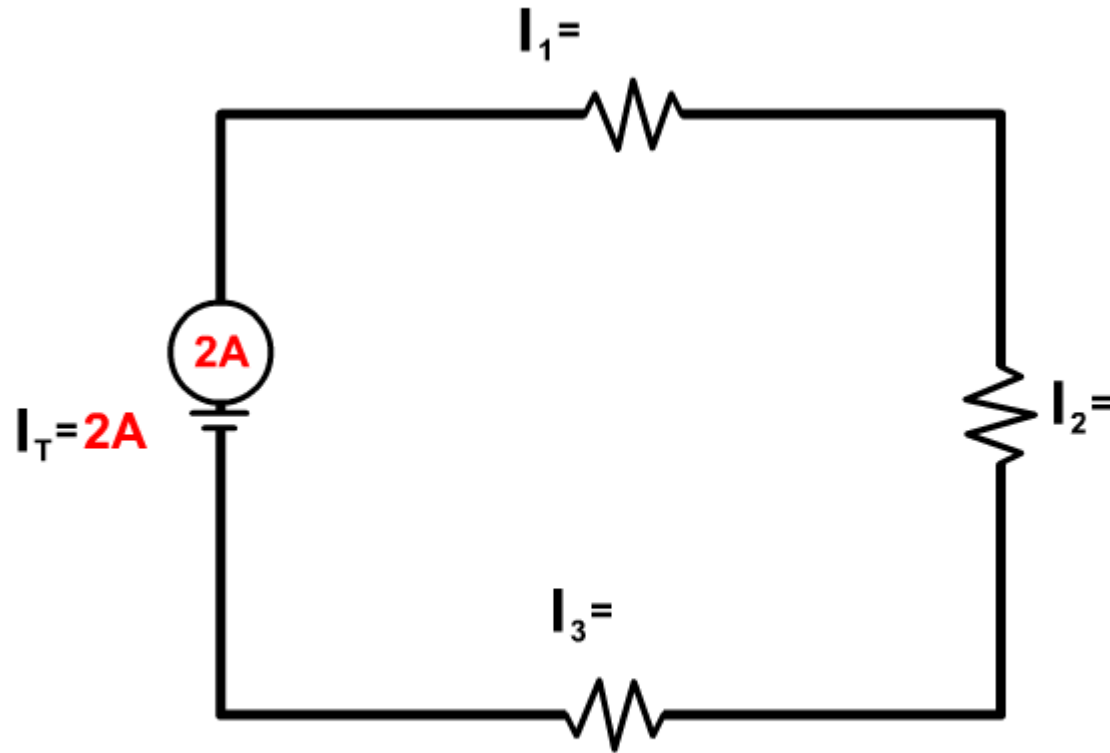
0 V



Series Circuits

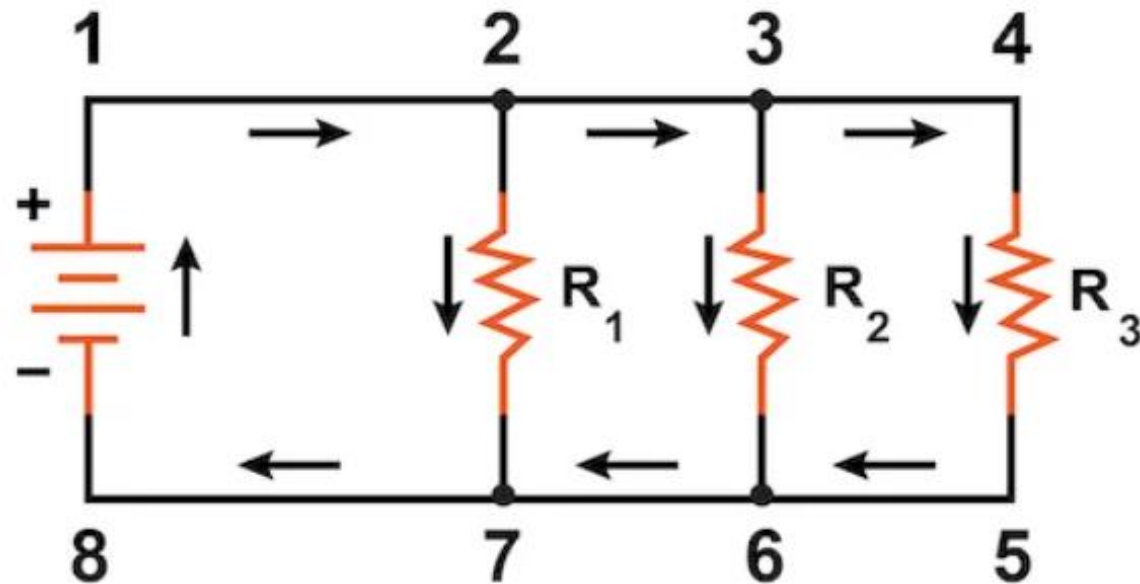


Series Circuits

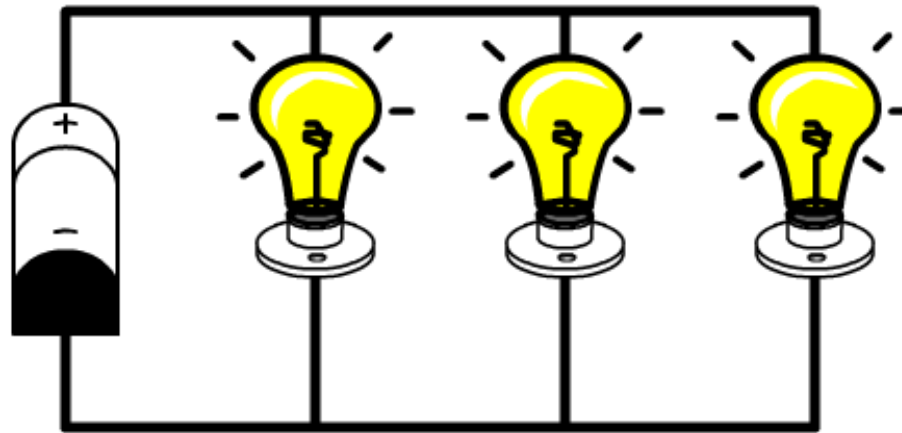


Series Circuit Current

Parallel Circuits



Parallel Circuits



Parallel Circuit

Parallel Circuits

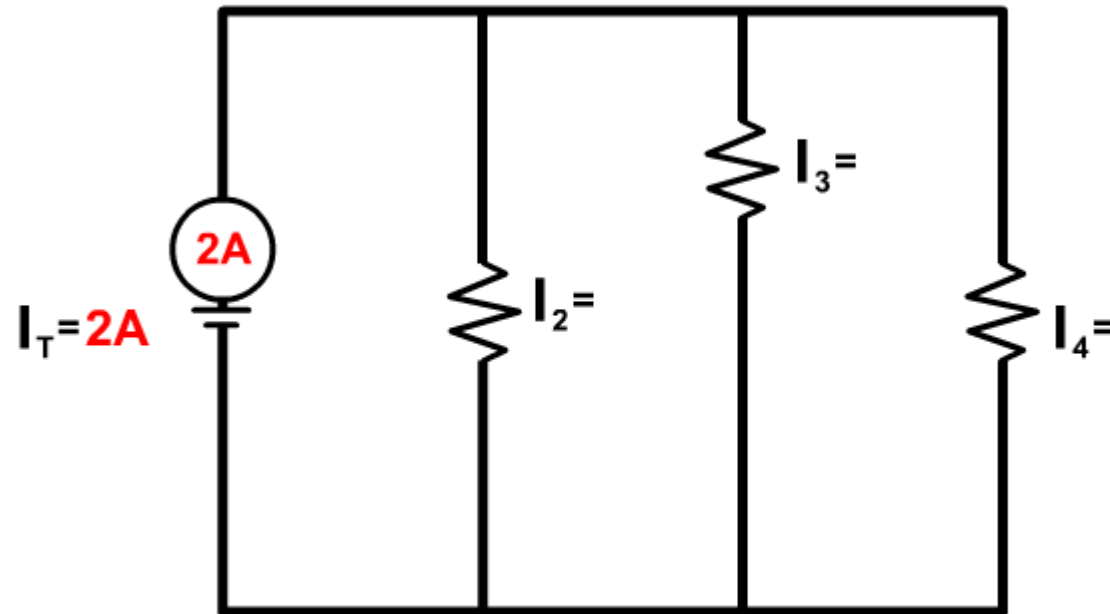
Parallel Circuit Rules

$$V_T = V_1 = V_2 = V_3 = \dots$$

$$I_T = I_1 + I_2 + I_3 + \dots$$

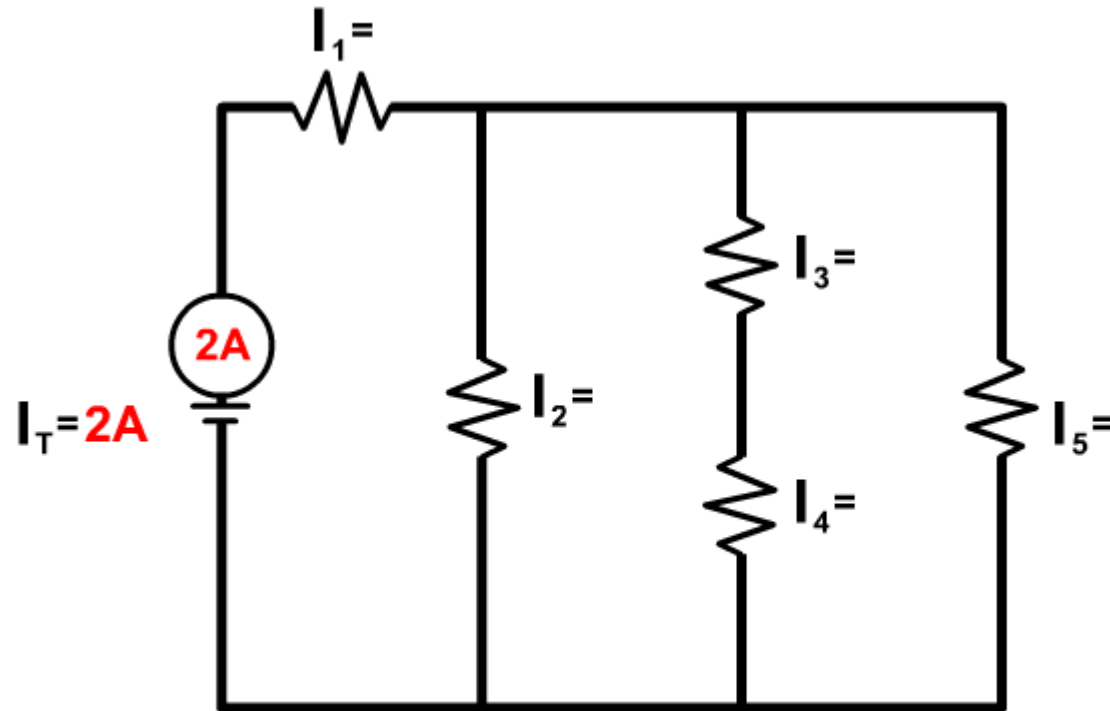
$$1/R_T = 1/R_1 + 1/R_2 + 1/R_3 + \dots$$

Parallel Circuits



Parallel Circuit Current

Series and Parallel Circuits



Complex Circuit Current

Thank you!



Questions

